ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (EE)

Semester: VI

SUBJECT CODE: BOE603

ANALOG AND DIGITAL COMMUNICATION

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (10 Marks)

Q. 1. Answer the following: (1 × 5 = 5 Marks)

(i) What do you understand by multiplexing?
(ii) What is the frequency deviation?
(iii) What is modem?
(iv) What is basic difference between analog and digital communication?
(v) What is the basic difference between TDM and FDM?

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) What do you mean by bandwidth of the signal?
(ii) Write the different channels used for communication systems.
(iii) What the optical communication?
(iv) What is the thermal noise?
(v) What is skip distance?

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) What is amplitude modulation? Derive an expression for current in a single-tone amplitude–modulated wave.

OR

(b) What is the role of modulator in the communication systems? Explain the balance modulator in detail.

(ii)(a) What do you mean by uplink and downlink? Explain the various type of satellite used in the communication.

OR

(b) What is the basic concept of angle modulation? Derive an expression for a single tone frequency modulated wave.

Q.4.(i) (a) What is multiplexing? Explain the various multiplexing techniques used in the communication systems.

OR

(b) An AM transmitter radiated 9Kwatts of power when the carrier is unmodulated and 10.125 Kwatt when the carrier is sinusoidally modulated. Find the modulation index, percentage modulation.

(ii)(a) Explain the concept of envelop detector used in the amplitude modulation.

OR

(b) Explain the various pulse modulation schemes used in the communication systems.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5. (a) What are the various modes of propagation? Explain every mode of propagation in detail.

OR

(b) What is MUF? Derive the expression of MUF for flat surface.

Q.6. (a) Explain the optical fibre communication in detail. Using block diagram.

OR

(b) What are the advantages and disadvantages of analog communication over digital communication? Explain in detail.